

ANSWER KEY

Q.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
A.	2	1	4	1	3	1	2	4	2	3	4	2	3	2	4	4	2	2	2	4
Q.	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40
A.	4	3	3	1	2	2	2	2	2	2	1	2	4	2	1	1	2	3	4	1
Q.	41	42	43	44	45	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60
A.	2	3	4	3	3	2	2	2	2	3	4	2	2	4	2	1	3	2	1	1
Q.	61	62	63	64	65	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80
A.	2	4	4	3	1	3	1	2	4	3	3	3	1	4	3	4	2	1	2	3
Q.	81	82	83	84	85	86	87	88	89	90	91	92	93	94	95	96	97	98	99	100
A.	3	3	4	4	1	4	1	2	3	3	4	4	2	1	3	1	4	3	3	2
Q.	101	102	103	104	105	106	107	108	109	110	111	112	113	114	115	116	117	118	119	120
A.	4	2	2	1	2	3	2	2	3	3	2	3	2	1	1	2	3	3	4	2
Q.	121	122	123	124	125	126	127	128	129	130										
A.	2	3	1	2	2	2	1	2	2	3										

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HINT – SHEET

1. **Ans (2)**

$$y = 3x + 5 \Rightarrow \frac{dy}{dt} = 3 \frac{dx}{dt}$$

$\Rightarrow$ y-coordinate changes at faster rate.

2. **Ans (1)**

$$A - B = 10$$

$$\sqrt{A^2 + B^2} = 50$$

$$\text{Solving } A = 40, B = 30$$

3. **Ans (4)**

$$T \propto v$$

$$R \propto u^2$$

4. **Ans (1)**

$$V_1 = V_2 \sin \theta$$

5. Ans (3)



$$\text{min force } F = \mu m_1 g + \frac{\mu m_2 g}{2}$$

6. Ans (1)

$$\frac{mv^2}{r} = \frac{k}{r^2} \Rightarrow mv^2 = \frac{k}{r} \therefore \text{K.E.} = \frac{1}{2} mv^2 = \frac{k}{2r}$$

$$\text{P.E.} = \int F dr = \int \frac{k}{r^2} dr = -\frac{k}{r}$$

$$\therefore \text{Total energy} = \text{K.E.} + \text{P.E.} = \frac{k}{2r} - \frac{k}{r} = -\frac{k}{2r}$$

7. Ans (2)

$$\text{K.E} = \frac{1}{2} mv^2 = As$$

$$\frac{dK}{dt} = vm \frac{dv}{dt} = A \frac{ds}{dt} \Rightarrow m \frac{dv}{dt} = A$$

$$dv = \frac{A}{m} dt \Rightarrow v = \frac{A}{m} t$$

$$a_t = \frac{A}{m}, a_c = \frac{v^2}{R} = \frac{A^2 t^2}{m^2 R}$$

$$a_0 = \sqrt{a_t^2 + a_c^2} = \sqrt{\frac{A^2}{m^2} + \left(\frac{A^2 t^2}{m^2 R}\right)^2}$$

$$\text{Also, } \frac{ds}{dt} = \frac{A}{m} t \Rightarrow s = \frac{A}{2m} t^2 \Rightarrow t^2 = \frac{2ms}{A}$$

$$\text{Then, } a = \sqrt{\frac{A^2}{m^2} + \left(\frac{A^2}{m^2 R} \times \frac{2ms}{A}\right)^2}$$

$$= \sqrt{\frac{A^2}{m^2} + \frac{4A^2 s^2}{m^2 R^2}} = \frac{A}{m} \sqrt{1 + \frac{4s^2}{R^2}}$$

$$\text{Thus, the net force on the particle} = ma = A \sqrt{1 + \frac{4s^2}{R^2}}$$

8. Ans (4)

$$\text{For equilibrium position, } \frac{du}{dx} = 0$$

$$\Rightarrow x^6 = \frac{2a}{b}$$

$$U(x = \infty) = 0$$

$$U(x = \text{equilibrium position})$$

$$= \frac{a}{\left(\frac{2a}{b}\right)^2} - \frac{b}{\left(\frac{2a}{b}\right)} = -\frac{b^2}{4a}$$

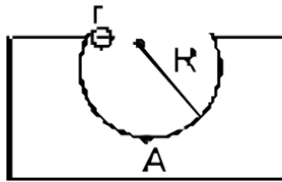
9. **Ans (2)**

when ball reaches pt A

then block get shifted by x

$\therefore$ but since there is no ext

force therefore com remain at its position



$$[(R-r) - x]m = Mx$$

$$\therefore x = \frac{m(R-r)}{M+m}$$

10. **Ans (3)**

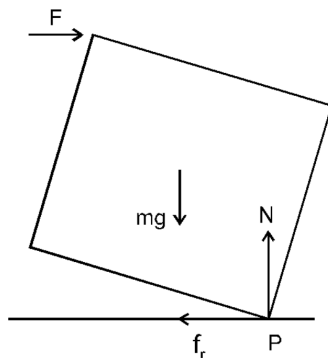
At the critical condition, normal reaction N will pass through point P .

In this condition

$$\tau_N = 0 = \tau_{fr} \text{ (About P)}$$

the block will topple when

$$\tau_F > \tau_{mg}$$



or

$$FL > (mg) \frac{L}{2}$$

$\therefore$

$$F > mg / 2$$

11. **Ans (4)**

$$\rho gh = \rho' gh'$$

$$h' = \frac{\rho h}{\rho'} = \frac{13.6 \times 70}{3.4} = 280 \text{ cm}$$

12. **Ans (2)**

$$g = \frac{4}{3} \pi \rho G R$$

If density is same then $g \propto R$

According to problem $R_p = 2R_e \therefore g_p = 2g_e$

For clock P (based on pendulum motion)

$$T = 2\pi \sqrt{\frac{\ell}{g}}$$

Time period decreases on planet so it will run faster because $g_p > g_e$

For clock S (based on oscillation of spring)

$$T = 2\pi \sqrt{\frac{m}{k}}$$

So it does not change

13. **Ans (3)**

$$Y = \frac{F/A}{\Delta \ell / \ell} \Rightarrow \Delta \ell = \frac{(F)(\ell)}{AY} = \frac{(10)(1.1)}{(10^{-6})(1.1 \times 10^{11})} = .1 \text{ mm}$$

14. **Ans (2)**

Heat given by water,

$$Q_1 = ms \Delta T = 200 \times 1 \times (25 - 10) = 3000 \text{ cal}$$

Heat absorbed by m gm of ice at -14°C to convert into water at 10°C is :

$$Q_2 = (ms \Delta T)_{\text{ice}} + mL_{\text{ice}} + (ms \Delta T)_{\text{water}} \\ = m[0.5 \times 14 + 80 + 1 \times 10] = 97m$$

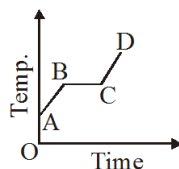
$$\text{Hence, } 97m = 3000 \text{ or } m = 31 \text{ gm}$$

15. **Ans (4)**

In isothermal process $\Delta Q \neq 0$.

16. **Ans (4)**

Liquid oxygen when heated from 50 K to 300 K, its phase changes from liquid to gaseous state. During this phase change the temperature of oxygen will remain constant and before and after the phase change the temperature increase linearly. Thus AB represents liquid state and CD gaseous state.



17. **Ans (2)**

$$\lambda_m T = b \text{ where } b = 2.89 \times 10^{-3} \text{ mK}$$

$$\Rightarrow T = \frac{b}{\lambda_m} = \frac{2.89 \times 10^{-3}}{1.5 \times 10^{-6}} \approx 2000 \text{ K}$$

18. **Ans (2)**

Radiation pressure

$$= \frac{2P}{cA} = \frac{2 \times 3 \times 10^{-3}}{3 \times 10^8 \times 3 \times 10^{-6}} \\ = 6.6 \times 10^{-6} \text{ N/m}^2$$

19. **Ans (2)**

Nuclear force is charge independent.

20. **Ans (4)**

$$\lambda = \frac{h}{mv} = \frac{h}{\sqrt{2mE}}$$

21. **Ans (4)**

$$g = \frac{GM}{R^2}$$

$$M = \text{constant } g \propto \frac{1}{R^2}$$

$$100 \frac{\Delta g}{g} = -2 \frac{\Delta R}{R} 100$$

$$\% \text{ change} = -2(-2)$$

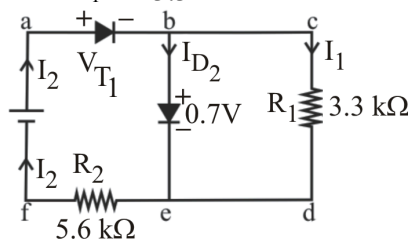
$$\% \text{ change in } g = 4\%$$

$$\text{increase by } 4\%$$

22. Ans (3)

The direction of conventional current is such as to turn both the diodes on. Resistor R_1 is parallel to diode D_2 ; hence

$$I_1 = \frac{V_{T_2}}{R_1} = \frac{0.7}{3.3} = 0.212 \text{ mA}$$



Applying Kirchhoff's voltage law around the loop abefa in clockwise direction,

$$-V_{T_1} - V_{T_2} - V_2 + E = 0$$

$$\text{So } V_2 = E - V_{T_1} - V_{T_2}$$

$$= 20 - 0.7 - 0.7 = 18.6 \text{ V}$$

$$\text{and } I_2 = \frac{V_2}{R_2} = \frac{18.6}{5.6} = 3.32 \text{ mA}$$

At the junction e,

$$I_2 = I_1 + I_{D_2}$$

$$\text{Hence, } I_{D_2} = I_2 - I_1$$

$$= 3.32 - 0.212 = 3.108 \text{ mA}$$

23. Ans (3)

$U_f = \text{same}$

$$W = \Delta U \Rightarrow \text{same work.}$$

24. Ans (1)

$$\frac{E}{B} = C, B = \frac{27}{3} \times 10^{-8} = 9 \times 10^{-8}$$

$$x = A \sin \left(2\pi ft - \frac{2\pi}{\lambda} x \right)$$

wave - x direction hence,

$$E \rightarrow Y \text{ then } B \rightarrow Z$$

$$\text{or } E \rightarrow Z \text{ then } B \rightarrow Y$$

25. Ans (2)

$$\text{For glass-water interface } {}_g\mu_w = \frac{\sin i}{\sin r} \dots\dots\dots (i)$$

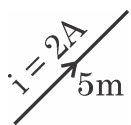
$$\text{For water-air interface } {}_w\mu_a = \frac{\sin r}{\sin 90^\circ} \dots\dots\dots (ii)$$

$$\Rightarrow {}_g\mu_w \times {}_w\mu_a = \frac{\sin i}{\sin r} \times \frac{\sin r}{\sin 90^\circ} = \sin i$$

$$\Rightarrow \frac{\mu_w}{\mu_g} \times \frac{\mu_a}{\mu_w} = \sin i \Rightarrow \mu_g = \frac{1}{\sin i}$$

26. **Ans (2)**

Joining end points.



$$F = i(I \times B)$$

$$2 \times 5 \times 2 = 20\text{N}$$

27. **Ans (2)**

$$\text{At } t = 0, \quad x = A, \quad x = A \cos \omega t$$

$$\text{After } t = \frac{T}{6} \quad x = A \cos \left(\frac{2\pi}{T}, \frac{T}{6} \right)$$

$$= A \cos 60^\circ = \frac{A}{2}$$

$$\therefore \text{kinetic energy } k = \frac{1}{2} m \omega^2 \left(A^2 - \frac{A^2}{4} \right)$$

$$\text{potential energy } U = \frac{1}{2} m \omega^2 \left(\frac{A}{4} \right)^2$$

$$\frac{k}{U} = \frac{3}{1}$$

28. **Ans (2)**

$$\text{General equation } y = f \left(t + \frac{x}{v} \right)$$

29. **Ans (2)**

$$\frac{nD\lambda_R}{d} = \frac{(n+1)D\lambda_B}{d}$$

30. **Ans (2)**

$$I = I_1 \sin \omega t + I_2 \cos \omega t$$

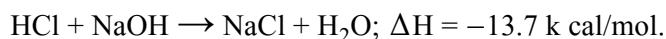
$$\therefore I_0 = \sqrt{I_1^2 + I_2^2}$$

$$\therefore I_{\text{rms}} = \frac{I_0}{\sqrt{2}} = \sqrt{\frac{I_1^2 + I_2^2}{2}}$$

31. **Ans (1)**

ZnCO_3 is not a sulphide ore.

32. **Ans (2)**



Heat of ionisation of CH_3COOH is

$$\Delta H = 13.7 - 3.7 = 10 \text{ kcal}$$

33. **Ans (4)**

DU

(A) 6

(B) 4

(C) 6

(D) 7

34. **Ans (2)**

SiO_2 is soluble in HF this reaction is known as etching of glass.

35. **Ans (1)**

Ans. (1)

36. **Ans (1)**

$$n_{\text{CaCl}_2} = \frac{5.5 \times 10^{-5}}{111} = 5 \times 10^{-5}$$

$$n_{\text{MgCl}_2} = \frac{9.5}{95} = 10^{-4}$$

For hardness in 1 L

$$n_{\text{CaCO}_3} = n_{\text{CaCl}_2} + n_{\text{MgCl}_2}$$

$$n_{\text{CaCO}_3} = 15 \times 10^{-5}$$

$$W_{\text{CaCO}_3} = 15 \times 10^{-3} \text{ gm}$$

$$(\text{Hardness in ppm}) = \frac{15 \times 10^{-3}}{10^3} \times 10^6 = 15 \text{ ppm}$$

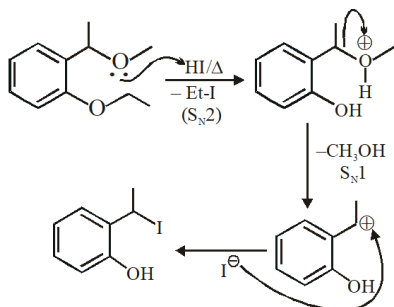
37. **Ans (2)**

Na_2Z -sodium zeolite.

$\text{Na}_2\text{Z} + \text{hard water} \rightarrow \text{CaZ on MgZ} + \text{Na}^+$

Ca^+ & Mg^{2+}

38. **Ans (3)**



It is nucleophilic substitution reaction.

39. **Ans (4)**

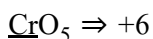
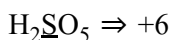
Antipyretic drug

40. **Ans (1)**

Permanganate ion (MnO_4^-) behaves as oxidizing agent in acidic medium with H_2O_2 .

41. **Ans (2)**

Oxidation number of

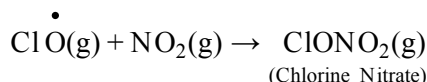


$$a + b + c = 6 + 6 + 3 = 15$$

42. **Ans (3)**

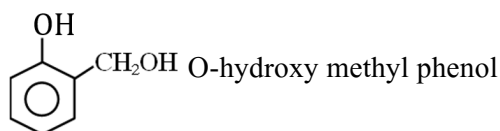
In electro-osmosis, the sol particles are prevented from migration whereas the dispersion medium migrates in the direction opposite to those of particles. Here, the medium is negatively charged.

43. **Ans (4)**



44. **Ans (3)**

Monomer of Novolac is



45. **Ans (3)**

$$\frac{1}{2}\text{N}_2 + \frac{3}{2}\text{H}_2 \rightarrow \text{NH}_3$$

$$\Delta H = \frac{1}{2}(\text{N}-\text{N}) + \frac{3}{2}(\text{H}-\text{H}) - 3(\text{N}-\text{H})$$

$$-46 = \frac{1}{2}712 + \frac{3}{2}(436) - 3(\text{N}-\text{H})$$

$$\text{N}-\text{H} = +352 \text{ kJ}$$

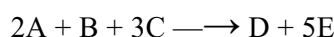
46. **Ans (2)**

$$\text{Basic nature of hydroxide} \propto \frac{1}{\text{EN of element}} \propto \frac{1}{Z_{\text{eff}}}$$

In lanthanide series due to lanthanide contraction size decreases & z_{eff} increases, and EN also increases so basicity of oxides and hydroxides decreases.

47. **Ans (2)**

For reaction



$$\text{ROR} = \frac{1}{2} \left[\frac{-d(\text{A})}{dt} \right] = \frac{1}{1} \left[\frac{-d(\text{B})}{dt} \right] = \frac{1}{3} \left[\frac{-d(\text{C})}{dt} \right] = \frac{1}{1} \left[\frac{d(\text{D})}{dt} \right] = \frac{1}{5} \left[\frac{d(\text{E})}{dt} \right]$$

We know $\text{ROR} = \text{K} \text{A}^x \text{B}^y \text{C}^z$ (A)

From exp. (1) $2.5 \times 10^{-4} = \text{K}(0.01)^x (0.01)^y (0.01)^z$ (1)

For exp. (2) $10^3 = \text{K}(0.02)^x (0.01)^y (0.01)^z$ (2)

For exp. (3) $2.5 \times 10^{-4} = \text{K}(0.01)^x (0.02)^y (0.01)^z$ (3)

For exp. (4) $5 \times 10^{-4} = \text{K}(0.01)^x (0.02)^y (0.02)^z$ (4)

Equation (3) / equation (1)

$$1 = 2^y \Rightarrow \boxed{y = 0}$$

Equation (4) / equation (3)

$$2 = 2^z \Rightarrow \boxed{z = 1}$$

Equation (2) / equation (1)

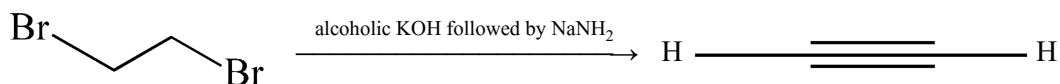
$$\frac{10^{-3}}{2.5 \times 10^{-4}} = 2^x$$

$$4 = 2^x \Rightarrow \boxed{x = 2}$$

Put the value x, y, z in equation (A)

$$\text{So ROR} = \text{k}[\text{A}]^2 [\text{B}]^0 [\text{C}]^1$$

48. Ans (2)

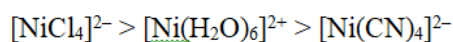


49. Ans (2)

- (1) $[\text{Cr}(\text{H}_2\text{O})_6] \text{Cl}_3 \rightarrow (4)$
 (2) $[\text{Cr}(\text{H}_2\text{O})_4\text{Cl}_2] \text{Cl} \cdot 2\text{H}_2\text{O} \rightarrow (2)$
 (3) $[\text{Cr}(\text{H}_2\text{O})_5\text{Cl}] \text{Cl} \cdot 2\text{H}_2\text{O} \rightarrow (3)$
 (4) $[\text{Cr}(\text{H}_2\text{O})_3 \text{Cl}_3] \cdot 3\text{H}_2\text{O} \rightarrow (0)$

conductivity $\propto$ no. of ion

50. Ans (3)

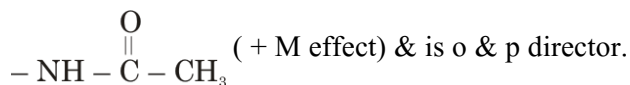


Splitting energy order $\Delta_t < \Delta_0 < \Delta_{sq}$

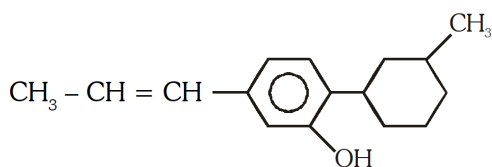
absorbed energy order $[\text{NiCl}_4]^{2-} < [\text{Ni}(\text{H}_2\text{O})_6]^{2+} < [\text{Ni}(\text{CN})_4]^{2-}$

intensity of colour of compound $[\text{NiCl}_4]^{2-} > [\text{Ni}(\text{H}_2\text{O})_6]^{2+} > [\text{Ni}(\text{CN})_4]^{2-}$

51. Ans (4)

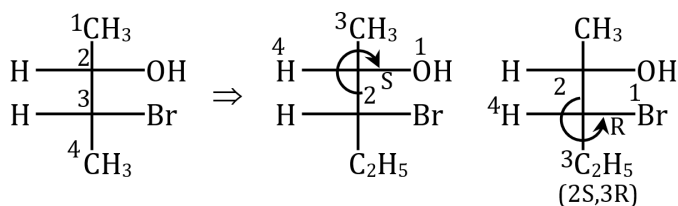


52. Ans (2)



Total Geometrical isomers = $2^2 = 4$

53. Ans (2)



54. Ans (4)

$$\text{PE} = \frac{kZe^2}{r}$$

for Li^{2+} ion : $z = 3$

$$\Rightarrow \text{P.E.} = -\frac{3ke^2}{r}$$

55. **Ans (2)**

$$(n + 1)$$

$$e_1 - 3s \quad 3$$

$$e_2 - 2p \quad 3$$

$$e_3 - 3p \quad 4$$

$$e_4 - 4s \quad 4$$

If the value of $(n + 1)$ is same lower the value of 'n' lower the energy

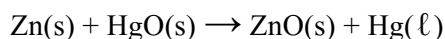
$$e_4 > e_3 > e_1 > e_2$$

56. **Ans (1)**

Because of low oxidation potential of Ag

it is get easily reduced. Means that

it is having higher reduction potential



57. **Ans (3)**

Clean water could have BOD value of less than

5 ppm whereas highly polluted water could have a BOD value of 17 ppm or more.

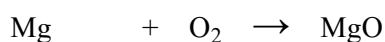
58. **Ans (2)**

$$\text{From : } \log \left(\frac{K_2}{K_1} \right) = \frac{\Delta H}{2.303R} \left[\frac{1}{T_1} - \frac{1}{T_2} \right]$$

$$\text{If } \Delta H = 0 \text{ then } \log \left(\frac{K_2}{K_1} \right) = 0$$

$$\text{i.e. : } K_2 = K_1$$

59. **Ans (1)**



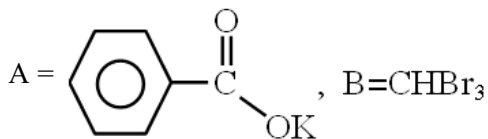
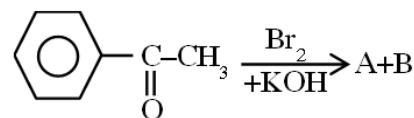
$$0.5 \text{ mole} \quad 1 \text{ mole}$$

L.R.

0.5 mole

$$\text{weight of MgO} = 0.5 \times 40 = 20 \text{ gm.}$$

60. **Ans (1)**



71. **Ans (3)**

Addition of first three digits and then last 2 digits i.e. $7 + 2 + 6 = 15$

$$4 + 8 = 12$$

i.e. 1512.

72. **Ans (3)**

First and third are both parts of the second.

73. **Ans (1)**

The upper element is converted to an element similar to the lower elements and each one of the lower elements is converted to an element similar to the upper element.



74. **Ans (4)**

By observation,



75. **Ans (3)**

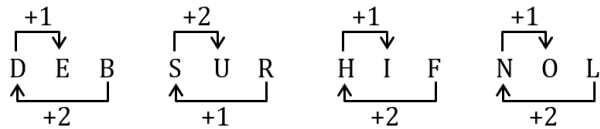
On close observation, we find that the option (3) will complete the pattern when placed in the blank space of question figure as shown below:



76. **Ans (4)**

Except option (4), in all others the ratio between two numbers is 8 : 7.

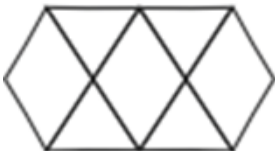
77. **Ans (2)**



78. **Ans (1)**

In all other figures, one-fourth part of the design is shaded. In figure (1) only one-third part is shaded.

79. **Ans (2)**



80. **Ans (3)**

By observation,



81. **Ans (3)**

The word formed is TECHNIQUE.

Thus, the missing letters are E, I, U.

82. **Ans (3)**

Sum of numbers vertically is square of a number i.e.

$$1 + 2 + 1 = 4 = 2^2$$

$$2 + 5 + 2 = 9 = 3^2$$

$$3 + 12 + 1 = 16 = 4^2$$

$$2 + 10 + 13 = 25 = 5^2$$

$$10 + 16 + 10 = 36 = 6^2$$

$$12 + 13 + 24 = 49 = 7^2$$

83. **Ans (4)**

Clearly,

$$1 + 1^3 = 2$$

$$2 + 2^2 = 6$$

$$6 + 3^3 = 33$$

$$33 + 4^2 = 49$$

$$49 + 5^3 = 174$$

$$174 + 6^2 = 210$$

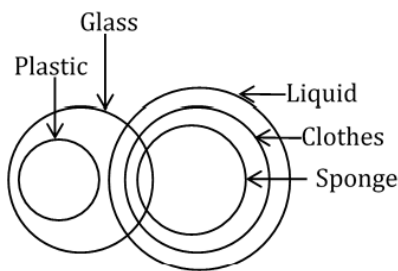
$$210 + 7^3 = 553$$

84. **Ans (4)**

By observation



85. **Ans (1)**



All sponges are clothes + All clothes are liquids = A + A = All sponges are liquids → conversion → Some liquids are sponges (I).

Hence, I does not follows. All plastics are glasses + conversion of Some sponges are glasses = A + I = No conclusion.

Hence, II does not follow.

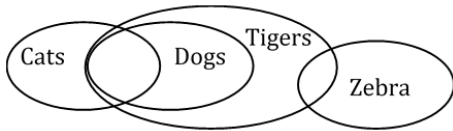
All plastics are glasses (1) → conversion → Some glasses are plastics (I).

Hence, III does not follow.

All clothes are liquids (1) conversion Some liquids are clothes (I).

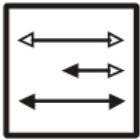
Hence, IV does not follow.

86. Ans (4)

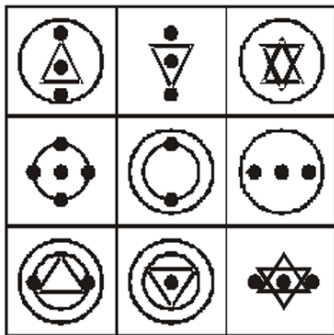


87. Ans (1)

By observation,



88. Ans (2)



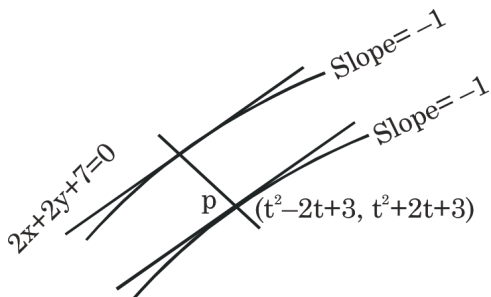
89. Ans (3)

Nothing about the comparison of pollution levels in India and the western world, can be deduced from the given passage.

90. Ans (3)

The fact is neither mentioned in nor can be derived from the passage.

91. Ans (4)



$$\frac{dy}{dx} = \frac{2t+2}{2t-2} = -1$$

$$\Rightarrow t = 0$$

Equation of line through point p

$$y - x = 0$$

Solve $y - x = 0$

$$\text{and } 2x + 2y + 7 = 0$$

92. **Ans (4)**

$$S_3 = \int_0^4 \frac{x^2}{4} dx$$

$$S_3 = \left[\frac{x^3}{12} \right]_0^4$$

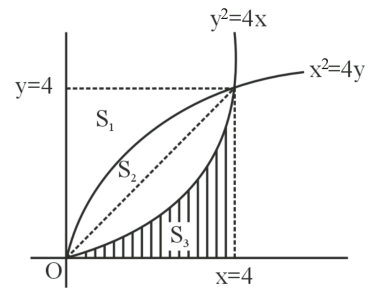
$$S_3 = \frac{4^3}{12} = \frac{16}{3}$$

$$S_1 = \frac{16}{3}$$

$$S_1 + S_2 + S_3 = 16$$

$$\therefore S_2 = \frac{16}{3}$$

$$\therefore S_1 : S_2 : S_3 = 1 : 1 : 1$$



93. **Ans (2)**

$$\Rightarrow \log_3 x = t^2 + 1$$

$$\therefore 2|t| - 3(1 + t^2) + 4 > 0$$

$$\therefore 3t^2 - 3|t| + |t| - 1 < 0$$

$$\Rightarrow |t| \in \left(-\frac{1}{3}, 1 \right) \Rightarrow t^2 \in [0, 1)$$

$$\Rightarrow x \in [3, 9) \Rightarrow 6 \text{ integers}$$

94. **Ans (1)**

$${}^{1000}C_0 + {}^{1000}C_1 x + \dots + {}^{1000}C_{1000} x^{1000} = (1 + x)^{1000}$$

$$\Rightarrow {}^{1000}C_1 + 2 {}^{1000}C_2 x + \dots + 1000 {}^{1000}C_{1000} x^{999} = 1000(1 + x)^{999}$$

$$\text{put } x = 1$$

$$\Rightarrow {}^{1000}C_1 + 2 {}^{1000}C_2 + \dots + 1000 {}^{1000}C_{1000}$$

$$= 1000 \times 2^{999}$$

95. **Ans (3)**

$$\max \left(x, \frac{1}{x} \right) = \begin{cases} x & ; \quad x \in [1, \infty) \\ \frac{1}{x} & ; \quad x \in (0, 1) \\ x & ; \quad x \in [-1, 0) \\ \frac{1}{x} & ; \quad x \in (-\infty, -1) \end{cases}$$

and

$$\min \left(x, \frac{1}{x} \right) = \begin{cases} \frac{1}{x} & ; \quad x \in [1, \infty) \\ x & ; \quad x \in (0, 1) \\ \frac{1}{x} & ; \quad x \in [-1, 0) \\ x & ; \quad x \in (-\infty, -1) \end{cases}$$

$$g(x) = \begin{cases} x^2 & ; \quad x \in [1, \infty) \\ \frac{1}{x^2} & ; \quad x \in (0, 1) \\ x^2 & ; \quad x \in [-1, 0) \\ \frac{1}{x^2} & ; \quad x \in (-\infty, -1) \\ 1 & ; \quad x = 0 \end{cases}$$

96. **Ans (1)**

$$f(3^-) = f(3^+)$$

$$\Rightarrow 2\alpha = 3\beta + 2$$

$$f'(3^-) = f'(3^+)$$

$$\Rightarrow \alpha = 4\beta$$

$$\alpha + \beta = 2$$

$$\{\alpha + \beta\} = 0$$

97. **Ans (4)**

Point of intersection is $(1, e^{-1})$

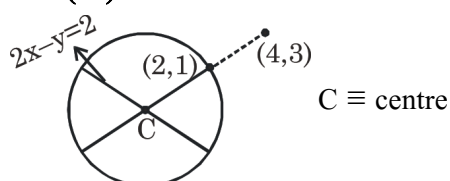
$$\frac{dy}{dx} = -e^{-|x|}$$

$$\text{Tangent : } y - \frac{1}{e} = -e^{-1}(x - 1)$$

$$-ey + 1 = x - 1$$

$$x + ey = 2$$

98. **Ans (3)**



Find equation of line passing through point $(4, 3)$ and $(2, 1)$ and solve this line with the given diameter.

99. Ans (3)

$$f(x) + f'(x) \leq 1$$

$$e^x f(x) + e^x f'(x) \leq e^x$$

$$\frac{d}{dx}(f(x) \cdot e^x) \leq e^x$$

$$\int_0^1 d(f(x) \cdot e^x) \leq \int_0^1 e^x dx$$

$$f(1) \cdot e^1 - f(0) \cdot e^0 \leq e^1 - 1$$

$$f(1) \leq 1 - \frac{1}{e}$$

100. Ans (2)

$$x dy - y dx + x \cos(\log_e x) dx = 0$$

$$\Rightarrow x^2 d\left(\frac{y}{x}\right) + x \cos(\log_e x) dx = 0$$

$$d\left(\frac{y}{x}\right) = \frac{-\cos(\log_e x)}{x}$$

$$\text{Integrate we get, } \frac{y}{x} = -\sin(\log_e x) + c$$

$$\frac{y}{x} = -\sin(\log_e x) + 1, y(e) = e(1 - \sin 1)$$

101. Ans (4)

$$\lim_{n \rightarrow \infty} \sum_{r=1}^{4n} \frac{1}{\sqrt{n}\sqrt{n+r}} = \int_0^4 \frac{dx}{\sqrt{1+x}} = 2(\sqrt{5} - 1)$$

102. Ans (2)

$$\int_{-2}^2 (ax^3 + cx) dx + \int_{-2}^2 (bx^2 + d) dx$$

↓

odd function

↓

even function

103. Ans (2)

Subtract (2) & (3) equations

$$(3 - \lambda)z = (10 - \mu)$$

(1) $\lambda \neq 3$ (Unique solution)

(2) $\lambda = 3, \mu = 10$ (infinite many solution)

(3) $\lambda = 3, \mu \neq 10$ (no solution)

104. Ans (1)

$$\therefore \sum a^2 + \sum ab \leq 0 \Rightarrow (a+b)^2 + (b+c)^2 + (c+a)^2 \leq 0$$

$$\Rightarrow a+b=0, b+c=0, c+a=0$$

$$\Rightarrow a=b=c=0$$

$$\therefore \begin{vmatrix} 4 & 0 & 1 \\ 1 & 4 & 0 \\ 0 & 1 & 4 \end{vmatrix} = 65$$

105. Ans (2)

$$(x-2)^2 + y^2 = \left(\frac{1}{\sqrt{2}}\right)^2 \left(\frac{x+y+1}{\sqrt{2}}\right)^2$$

$$\Rightarrow e = \frac{1}{\sqrt{2}}$$

106. Ans (3)

$$\frac{\pi}{2} \geq \sin^{-1}(\log_2 x) \geq 0$$

$$1 \geq \log_2 x \geq 0 \Rightarrow 1 \leq x \leq 2$$

107. Ans (2)

$$f(g(x)) = 3 + \tan x, \quad x \in \left[0, \frac{\pi}{4}\right]$$

$$0 \leq x \leq \frac{\pi}{4}$$

$$0 \leq \tan x \leq 1 \Rightarrow 3 \leq 3 + \tan x \leq 4$$

108. Ans (2)

$$\text{Use } T = S_1$$

109. Ans (3)

$$\frac{x^2}{12} - \frac{y^2}{7} = 1$$

$$y = mx \pm \sqrt{12m^2 - 7} \quad \text{passes through } (1, 2)$$

$$(2 - m)^2 = 12m^2 - 7$$

$$\Rightarrow 11m^2 + 4m - 11 = 0$$

$$m_1 m_2 = -1 \Rightarrow \theta = \frac{\pi}{2}$$

110. Ans (3)

Multiply N^r & D^r by x^2

$$\int \frac{x^5 + x^4 + x^3}{\sqrt{10x^6 + 12x^5 + 15x^4}} dx$$

put $10x^6 + 12x^5 + 15x^4 = t^2$

$$\int \frac{dt}{30} = \frac{t}{30} + c = \frac{x^2}{30} \sqrt{15 + 12x + 10x^2} + c$$

$$g(x) = x^2$$

111. Ans (2)

$$\int e^{x^2+x} (x(2x+1) + 1) \cdot dx$$

Using $\int e^{f(x)} \cdot [g(x) \times f'(x) + g'(x)] dx = g(x) \cdot e^{f(x)} + c$

So, integral is $\phi(x) = x \cdot e^{x^2+x} + c$

So, $f(x) = xe^x$

$f'(x) = 0$ at $x = -1$

$f''(x) > 0$ at $x = -1$

Hence minimum value is $m = -\frac{1}{e}$

$$\left[-\frac{1}{m}\right] = e = [2.7] = 2$$

112. **Ans (3)**

Possible only when $x = 0$

$$k = 1/2$$

113. **Ans (2)**

$$\int \frac{6x^{10} + 4}{x^5 \sqrt{x^6 - 3 - x^{-4}}} dx = \int \frac{6x^5 + 4x^{-5}}{\sqrt{x^6 - 3 - x^{-4}}} dx$$

$$x^6 - 3 - x^{-4} = t$$

$$(6x^5 + 4x^{-5}) dx = dt$$

114. **Ans (1)**

$$\lim_{x \rightarrow 0} \frac{f(x)}{g(x)} \cdot \frac{(1 - \cos x)}{\tan^2 x} = \frac{3}{7} \Rightarrow \lim_{x \rightarrow 0} \frac{f(x)}{g(x)} = \frac{6}{7}$$

$$e^{\frac{f(x)}{g(x)}} = e^{\frac{6}{7}}$$

115. **Ans (1)**

p : The teacher teaches well

q : doubt decreases

Negation of $(p \rightarrow q) \equiv p \wedge \sim q$

The teacher teaches well and doubt doesn't decrease.

116. **Ans (2)**

$$\text{adj} A = |A| A^{-1}$$

$$(\text{adj}(A^{-1}))^{-1} = (|A^{-1}| (A^{-1})^{-1})^{-1}$$

$$(\text{adj}(A^{-1}))^{-1} = \left(\frac{A}{|A|} \right)^{-1}$$

$$(\text{adj}(A^{-1}))^{-1} = |A| A^{-1}$$

$$|(\text{adj}(A^{-1}))^{-1}| = ||A| A^{-1}|$$

$$= |A|^3 \times \frac{1}{|A|} = |A|^2$$

117. **Ans (3)**

Use $A^6 = A^5 = A^4 = A^3 = A^2 = A$

and take determinant on both side

118. **Ans (3)**

Possible set of three consecutive numbers are $\{1, 2, 3\}$, $\{2, 3, 4\}$, $\{3, 4, 5\}$, ..., $\{48, 49, 50\}$. which are 48 in number.

$$\text{Thus, required probability} = \frac{{}^{48}C_1}{{}^{50}C_3} = \frac{3}{(25)(49)}$$

119. **Ans (4)**

$${}^6C_1 \cdot \frac{1}{2} \left[{}^5C_1 \cdot \left(\frac{1}{6} \right)^4 \left(\frac{1}{3} \right) + \left(\frac{1}{6} \right)^5 \right] \frac{1}{2} = \frac{11}{2592} \times \frac{1}{2}$$

120. **Ans (2)**

$${}^nC_2 C_2 = \frac{{}^{n(n-1)}{C_2}}{2} = \frac{\frac{n(n-1)}{2} \times \left(\frac{n(n-1)}{2} - 1 \right)}{2}$$

$$= \frac{1}{8} n(n-1)(n-2)(n+1) = 3 {}^{n+1}C_4$$

121. Ans (2)

The required no of ways

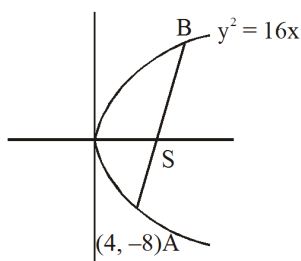
$$\begin{aligned}
 &= {}^{10}C_3 + {}^{10}C_4 + {}^{10}C_5 + \dots + {}^{10}C_6 + {}^{10}C_7 \\
 &= 2^{10} - ({}^{10}C_0 + {}^{10}C_1 + {}^{10}C_2 + {}^{10}C_8 + {}^{10}C_9 + {}^{10}C_{10}) \\
 &= 2^{10} - 2(1 + 10 + 45) = 912
 \end{aligned}$$

122. Ans (3)

$$\begin{aligned}
 y &= mx + \frac{64}{m}, \quad y = mx - 8m^2 \\
 \Rightarrow \frac{64}{m} &= -8m^2 \Rightarrow m = -2 \\
 \Rightarrow y \text{ int.} &= -32
 \end{aligned}$$

123. Ans (1)

Method-1



As AB is focal chord

$$t_1 t_2 = -1$$

$$(at_1^2, 2at_1) \equiv (4, -8)$$

$$2at_1 = -8 \Rightarrow t_1 = \frac{-8}{2a} = \frac{-8}{2 \times 4}$$

$$t_1 = -1$$

$$\text{So } t_2 = +1$$

$$\Rightarrow (at_2^2, 2at_2) = (4(1)^2, 2 \times 4 \times (1)) = (4, +8)$$

tangent at B:

$$(+8)y = 8(x + 4)$$

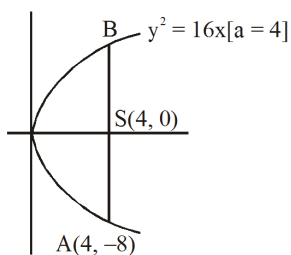
$$y = x + 4$$

Method-2

When we draw the graph of following parabola. We can see that the point A(4, -8) is one end of latus rectum.

So the other end of this focal chord will be other end of latus rectum, which is B(4, 8)

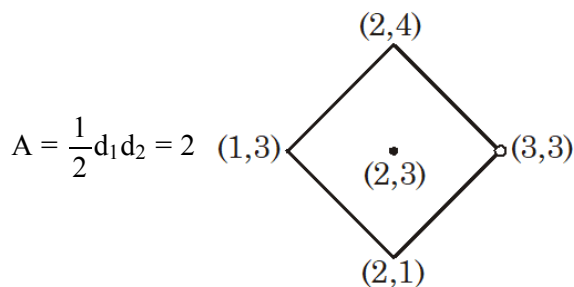
Equation of tangent at B(4, 8) is



$$(8)y = 8(x + y)$$

$$y = x + 4$$

124. Ans (2)



125. Ans (2)

$$f(x) = x^2 - (a+1)x + a - 1 = 0$$

$$\Rightarrow (x-a)(x-1) = 1.$$

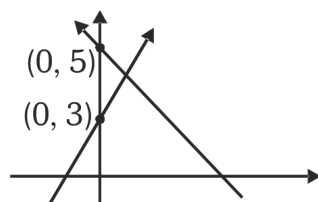
$$a \in \mathbb{Z}, x \in \mathbb{Z} \Rightarrow x-a=1, x-1=1 \text{ or}$$

$$x-a=-1, x-1=-1$$

$$\Rightarrow x=2, a=1 \text{ or } x=0, a=1$$

$$\Rightarrow a=1 \text{ is the only integral value of } a.$$

126. Ans (2)



$$ax + x = 2$$

$$(a+1)x = 2$$

$$a = 0 \text{ \& } a = 1$$

127. Ans (1)

by fundamental principal of counting 2^{n^2-n} .

128. Ans (2)

As $x, y \in \mathbb{R}$ and $xy > 0$, so x and y will be of same sign.

All the quantities $\frac{2x}{y^3}, \frac{x^3y}{3}, \frac{4y^2}{9x^4}$ are positive.

A.M. $\geq$ G.M.

$$\Rightarrow \frac{2x}{y^3} + \frac{x^3y}{3} + \frac{4y^2}{9x^4} \geq 3 \left(\left(\frac{2x}{y^3} \right) \left(\frac{x^3y}{3} \right) \left(\frac{4y^2}{9x^4} \right) \right)^{\frac{1}{3}}$$

$$= 3 \times \frac{2}{3} = 2$$

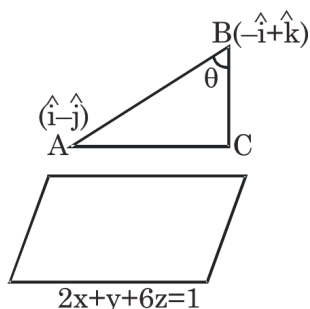
129. Ans (2)

$$AC = AB \sin \theta,$$

$$\vec{AB} = 2\hat{i} - \hat{j} - \hat{k}$$

$$\vec{n} = 2\hat{i} + \hat{j} + 6\hat{k}$$

$$\cos \theta = \frac{\vec{AB} \cdot \vec{n}}{|\vec{AB}| \cdot |\vec{n}|} = \frac{3}{\sqrt{6}\sqrt{41}}$$



$$AC = \sqrt{6} \sqrt{1 - \frac{9}{6 \times 41}} \\ = \sqrt{6} \sqrt{1 - \frac{3}{82}} = \sqrt{\frac{237}{41}}$$

130. Ans (3)

$$L_1 : \vec{r} = (2 + \lambda)\hat{i} + (9 + 2\lambda)\hat{j} + (13 + 3\lambda)\hat{k}$$

$$L_2 : \vec{r} = (-3 - \mu)\hat{i} + (7 + 2\mu)\hat{j} + (p - 3\mu)\hat{k}$$

for point of intersection

$$2 + \lambda = -3 - \mu \quad \dots(1)$$

$$9 + 2\lambda = 7 + 2\mu \quad \dots(2)$$

$$13 + 3\lambda = p - 3\mu \quad \dots(3)$$

Solving we get

$$\lambda = -3$$

$$\mu = -2$$

$$p = -2$$